

*The Friendly Guide to*

# What's Happening Inside an AI's Brain



*(explained so a 5-year-old gets it,  
but an expert learns something too)*

Mechanistic Interpretability with Sparse Autoencoders

CS 463 | University of San Francisco | Spring 2026

● math

● code

● facts

● creative

● emotions

● reasoning

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## Chapter 1

# What Is an AI?

### *The Word-Guessing Robot*

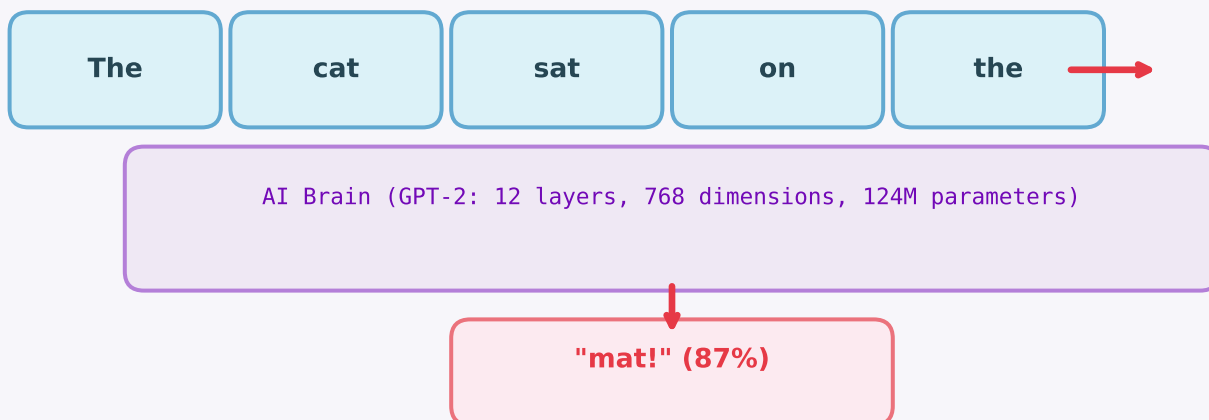
#### **If I were 5...**

Imagine you have a robot friend who is REALLY good at one game: you say the beginning of a sentence, and the robot guesses what word comes next. You say "The cat sat on the..." and the robot says "mat!" That's basically what an AI language model does. It reads your words and guesses the next one, over and over, until it writes a whole answer.

#### **The real deal:**

A Large Language Model (LLM) like GPT-2 is an autoregressive transformer that predicts  $P(\text{next\_token} \mid \text{previous\_tokens})$ . It has 124M parameters organized in 12 layers, each containing multi-head attention (12 heads) and an MLP. The model processes text as tokens (subwords) through a 768-dimensional residual stream.

#### **How the Word-Guessing Game Works**



## Chapter 2

# Layers

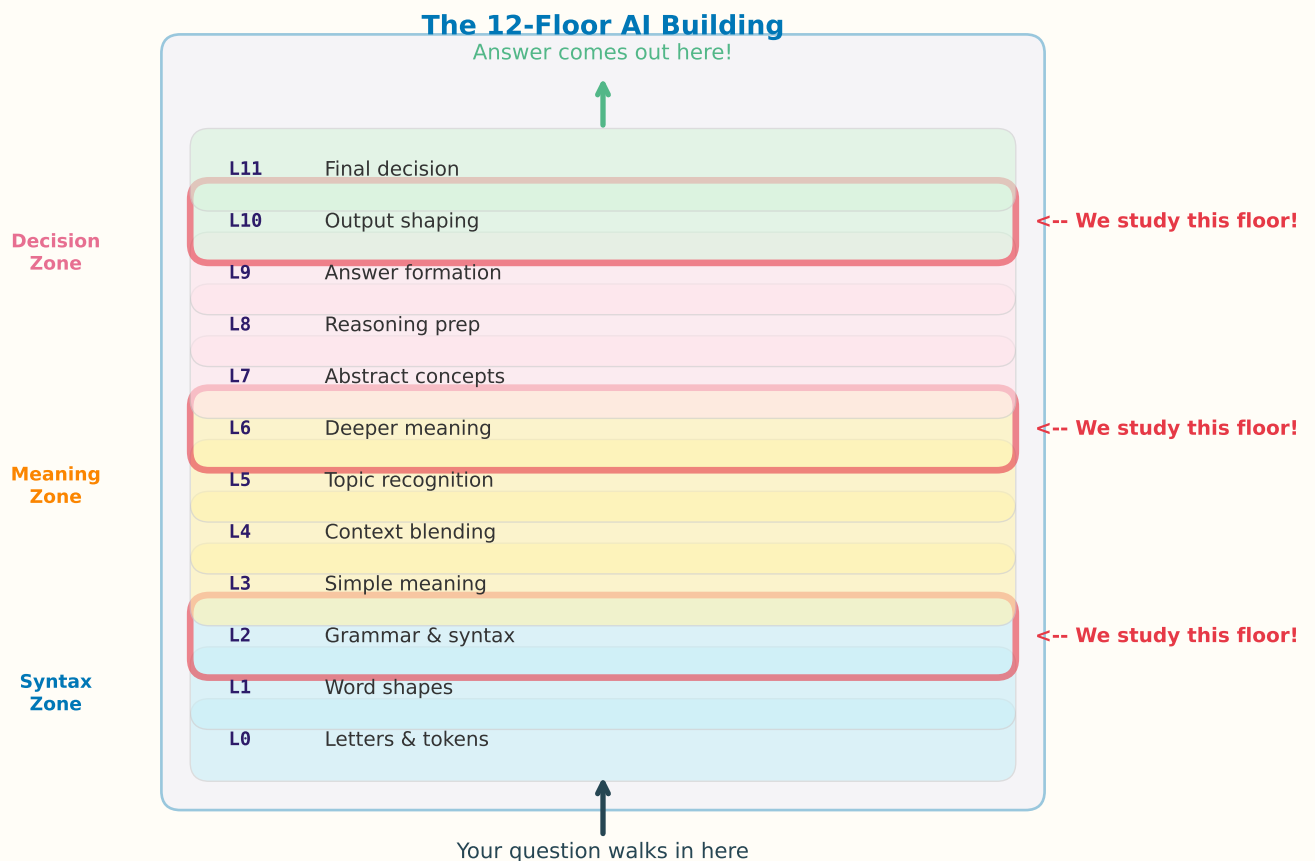
### *Floors in a Tall Building*

#### **If I were 5...**

The robot's brain is like a tall building with 12 floors. When you ask a question, your words walk in through the front door (floor 1) and climb up, floor by floor. On each floor, the words get changed a little bit -- they pick up new ideas. By the time they reach the top floor (floor 12), they've become the answer! The bottom floors figure out basic stuff ("this is English"), the middle floors figure out meaning ("this is a math question"), and the top floors decide what to say.

#### **The real deal:**

Each of GPT-2's 12 layers consists of multi-head self-attention followed by an MLP, with residual connections. The residual stream is the central 768-dim vector that accumulates information layer by layer. Early layers (0-3) handle syntax and token identity; middle layers (4-8) develop semantic representations; late layers (9-11) prepare the final next-token distribution via the unembedding matrix.



## Chapter 3

# Superposition

*Too Many Toys in One Box*

### **If I were 5...**

Imagine you have a toy box with only 5 slots, but you need to store 100 different toys. What do you do? You get clever! You stack the teddy bear AND the puzzle piece in the same slot. You squeeze the race car AND the doll together. Each slot now holds a MIXTURE of toys. If someone asks "what's in slot 3?" you can't give a simple answer -- it's a jumble. That's superposition: the AI's brain has 768 slots but needs to remember THOUSANDS of ideas, so it stuffs multiple ideas into each slot.

### **The real deal:**

Superposition occurs when a neural network represents more features (concepts) than it has dimensions. GPT-2 has `d_model=768`, but encodes thousands of distinct concepts. Features are stored as nearly-orthogonal directions in a high-dimensional space. Individual neurons become polysemantic -- each neuron responds to multiple unrelated concepts. This makes naive per-neuron interpretation unreliable. (Elhage et al., 'Toy Models of Superposition', 2022)

### **The Toy Box Problem**

#### **Easy: 5 toys, 5 slots**



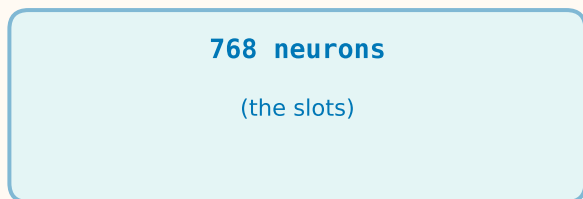
Each slot = 1 toy!

#### **Hard: 20 toys, 5 slots!**

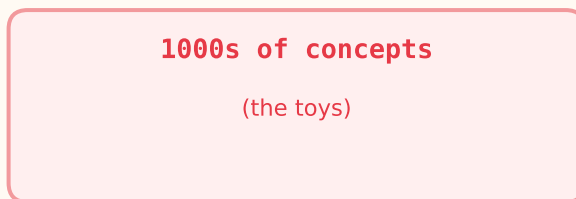


Each slot = a JUMBLE!

### **This is what happens in GPT-2's brain:**



must hold



## Chapter 4

# Sparse Autoencoders

*The Magic Sorting Machine*

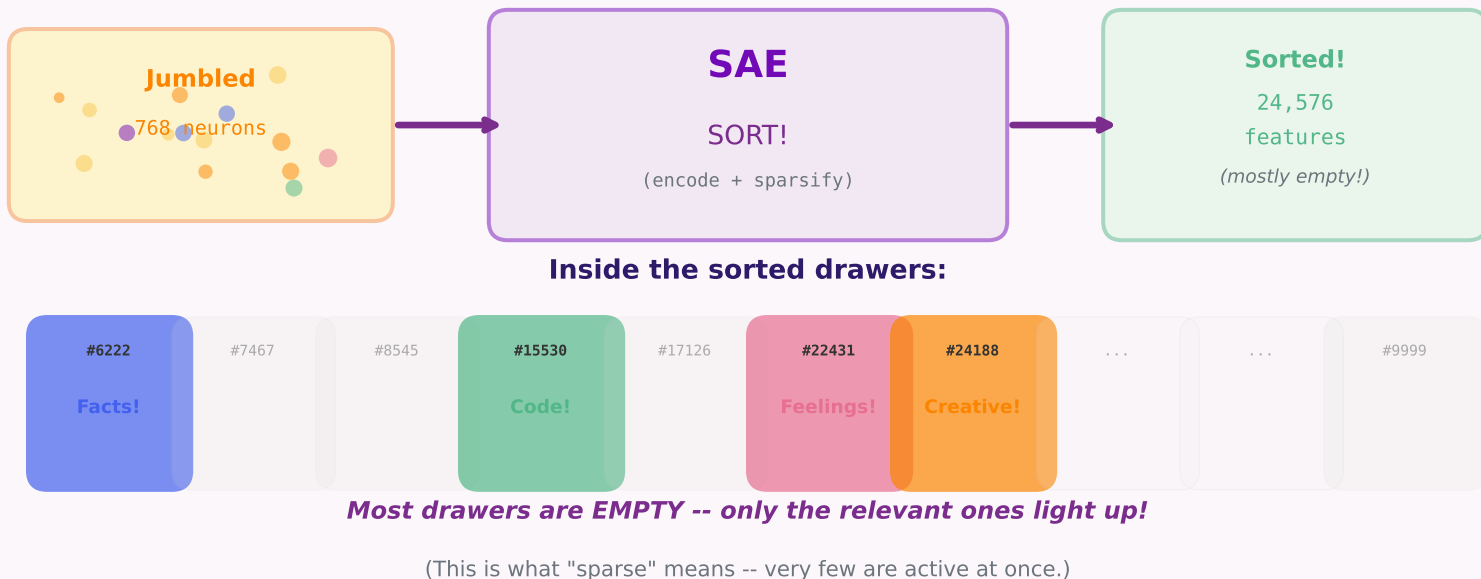
### If I were 5...

Remember the messy toy box where everything was jumbled together? Now imagine you have a MAGIC SORTING MACHINE. You dump in the jumbled mess, and out comes 24,576 tiny labeled drawers -- and each drawer has EXACTLY ONE toy in it! Drawer #15530 has the "code toy", drawer #6222 has the "facts toy", and so on. Most drawers are empty (the machine only opens the ones that matter for your question). This magic sorter is called a Sparse Autoencoder!

### The real deal:

An SAE is trained to reconstruct the 768-dim residual stream activation through a bottleneck:  $\text{encode}(x) = \text{ReLU}(W_{\text{enc}} @ x + b_{\text{enc}})$  producing a 24,576-dim sparse vector, then  $\text{decode}(z) = W_{\text{dec}} @ z + b_{\text{dec}}$  reconstructing the original 768 dims. The sparsity penalty (L1) ensures most of the 24,576 features are zero for any given input -- typically only 50-300 are nonzero. Each feature ideally corresponds to one monosemantic concept. We use the gpt2-small-res-jb release by Joseph Bloom (24,576 features per layer).

### The Magic Sorting Machine (SAE)



# Features

*The Individual LEGO Bricks*

## If I were 5...

You know how a LEGO castle is built from hundreds of tiny bricks? Each brick is simple -- a red 2x4, a blue 1x2, a green flat piece. But together they make something complex. Features are like those bricks! Feature #6222 might mean "the AI is thinking about a fact." Feature #15530 might mean "the AI is thinking about computer code." Each feature is one simple idea. When you ask the AI a question, different bricks (features) light up depending on what kind of question it is!

## The real deal:

Each of the 24,576 SAE features is a learned direction in the residual stream space. When a feature 'fires' (nonzero activation), it means the input triggered that particular concept. Features are monosemantic by design -- the L1 penalty during SAE training encourages each feature to respond to one thing. Our analysis identifies discriminative features: those that fire selectively for one prompt category. E.g., feature #6222 has a selectivity score of 10.25 for factual prompts -- the strongest single-category signal in our entire study.

## What Features Look Like (Layer 6 -- the middle floor)

### Feature #6222 "Facts Brick"

Fires when AI thinks about facts.  
"What is the capital of France?"

10.2

### Feature #15530 "Code Brick"

Fires when AI thinks about code.  
"Write a function to sort a list"

6.9

### Feature #22431 "Feelings Brick"

Fires when AI processes emotions.  
"I feel overwhelmed by everything"

5.7

### Feature #24188 "Creativity Brick"

Fires when AI thinks creatively.  
"Write a haiku about autumn"

5.6

*The longer the bar, the MORE selective that feature is for its category.*

## Chapter 6

# Our Experiment

Asking 120 Different Questions

### If I were 5...

We played a big game of "20 Questions" -- but instead of 20, we asked the robot 120 questions! And we split them into 6 types: math problems, coding puzzles, trivia facts, creative writing prompts, emotional feelings, and brain teasers. Then we peeked inside the robot's brain while it was thinking about each question, to see which drawers lit up. Would the math drawers light up for math? Would the feelings drawers light up for emotions?

### The real deal:

We curated 120 natural-language prompts: 20 per category across math, code, factual, creative, emotional, and reasoning. For each prompt, we ran the model forward, extracted the last-token residual stream at layers 2, 6, and 10, and passed it through the corresponding SAE to get a 24,576-dim feature activation vector. Total: 120 prompts x 3 layers = 360 feature vectors.

## The 6 Types of Questions We Asked

### 1 MATH

"What is 7 x 8?"

20 questions

### 2 CODE

"Sort this list"

20 questions

### 3 FACTS

"Capital of France?"

20 questions

### 4 CREATIVE

"Write a haiku"

20 questions

### 5 EMOTIONAL

"I feel overwhelmed"

20 questions

### 6 REASONING

"If all dogs are pets..."

20 questions

**Total: 120 prompts --> 360 brain scans (3 floors each!)**



## Chapter 7

# Feature Extraction

### Watching the Lights

#### If I were 5...

Now we watch carefully! We send each question into the robot's brain and put on special glasses (the SAE) that let us see which drawers light up. It's like an X-ray for robots! For each question, we check 3 floors (floor 2, floor 6, and floor 10) and write down every drawer that glows. Most drawers stay dark -- only a few dozen light up per question.

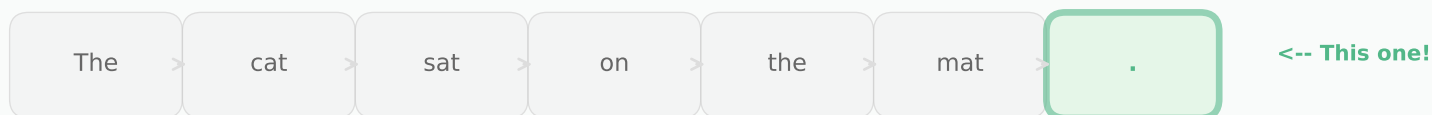
#### The real deal:

Extraction pipeline: (1) Tokenize prompt. (2) Forward pass with `transformer_lens` caching at `hook_resid_pre` for layers [2,6,10]. (3) Take last-token residual vector (shape [768]). (4) `SAE.encode(resid)` --> sparse feature vector (shape [24,576]). (5) Store as numpy array. The 'last token' is key: in causal attention, it aggregates information from all previous tokens, making it the richest representation of the full prompt.

### How We Scan the Robot's Brain



### Why the LAST token?



The last token has heard everything before it (causal attention).

*It's like the student who sat through the entire lecture -- they know the most!*

Output: 120 prompts x 3 layers = 360 brain scans, each with 24,576 features

## Chapter 8

# Sparsity

### How Many Lights Turn On?

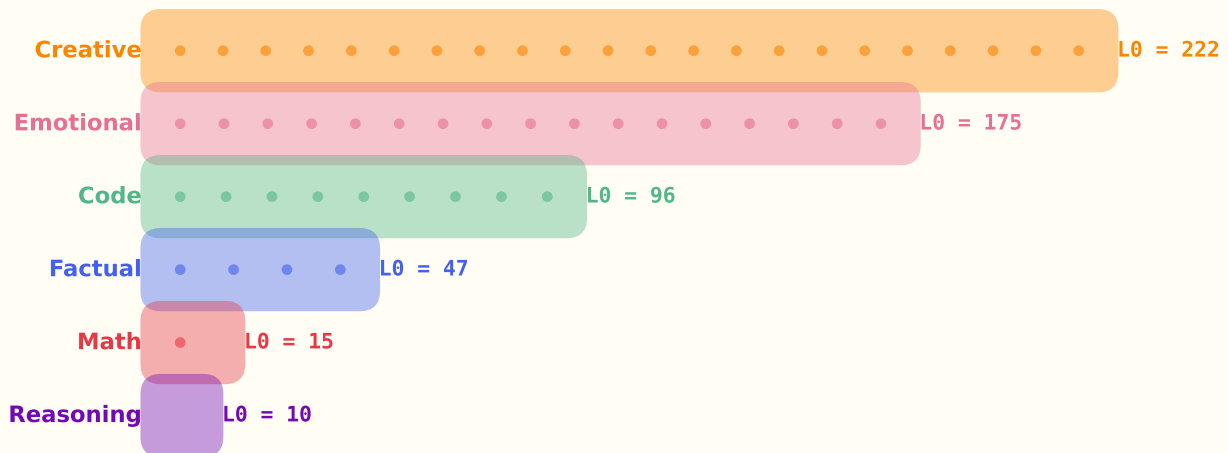
#### If I were 5...

Here's a surprising thing: out of 24,576 drawers, only about 50 to 300 actually light up for any question! That's like having a HUGE Christmas tree with 24,576 light bulbs, but only a handful turn on at a time. AND -- different types of questions turn on different AMOUNTS of lights! Creative writing questions turn on LOTS of lights (332!), like a big colorful party. Math questions only turn on a few (15), like a quiet reading lamp. We call this L0 -- it just counts how many lights are on.

#### The real deal:

L0 (the number of nonzero SAE features) varies dramatically by category. At layer 2: creative L0~221.8, emotional~175, code~96, factual~47, math~15.1, reasoning~9.9. At layer 10, all categories increase: creative hits 332.4. This reflects the representational complexity of each task -- creative writing requires activating diverse conceptual features (metaphor, imagery, emotion, rhythm), while math activates few but precise features.

### How Many Lights Turn On? (Layer 2)



#### What does this mean?

Creative writing needs MANY ideas at once (imagery + emotion + style + rhythm).  
Math needs just a FEW precise tools (numbers + operations). Broader tasks = more lights!

## Chapter 9

# Similarity & Clustering

*Which Questions Look Alike Inside?*

### **If I were 5...**

If you look at the light patterns for two math questions, they'd look very SIMILAR -- almost the same drawers light up! But if you compare a math question to a poem question, the patterns look VERY different. It's like how two cats look similar to each other, but very different from a spaceship. We measured this by asking: "How much overlap is there between the lights for each type of question?" This is called cosine similarity.

### **The real deal:**

We compute the mean feature activation vector for each category and measure pairwise cosine similarity. At layer 2, creative prompts show the lowest similarity to all other categories, while math/reasoning are moderately similar. By layer 10, categories converge (higher similarity), suggesting the model transitions from input-type-specific representations to a more uniform format for next-token prediction. We also use UMAP to project the 24,576-dim feature space into 2D for visual clustering.

## Do Similar Questions Have Similar Brain Patterns?

### **Math vs. Math**

*(very similar patterns!)*



"What is 3+5?"



**Same drawers light up!**

### **Math vs. Creative**

*(very different patterns!)*



"What is 7x8?"



**Different drawers light up!**

Cosine similarity measures how much two light patterns overlap.

High similarity = same type of thinking. Low similarity = totally different mental mode.

# Separability

*Can We Tell Them Apart?*

## If I were 5...

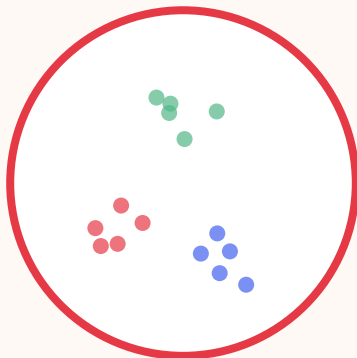
If you showed me two crayon drawings -- one by a kid who loves red and one by a kid who loves blue -- I could easily tell whose is whose! But if both kids use ALL the colors, it's harder to tell. Separability asks: "At which floor of the building is it EASIEST to tell the question types apart?" The answer surprised us: floor 2 (the BOTTOM)! The robot figures out "this is a math question" very early, then mixes everything together as it goes up.

## The real deal:

Fisher's discriminant ratio = between-class variance / within-class variance. A higher ratio means features better separate categories. Results: Layer 2 = 1.33 (best), Layer 6 = ~0.6, Layer 10 = ~0.6. This is counterintuitive -- one might expect deeper (more 'semantic') layers to be more discriminative. But it suggests rapid input categorization in early layers, followed by progressive transformation toward a universal generation-ready format.

## Where Is It Easiest to Tell Question Types Apart?

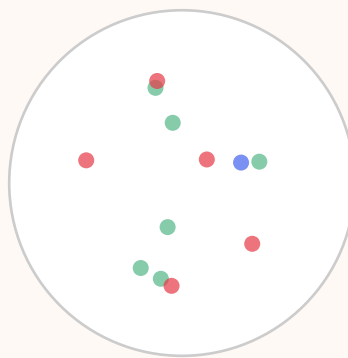
**Floor 2  
(early)**



Score: 1.33

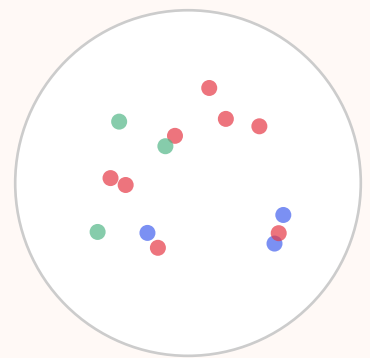
**BEST!**

**Floor 6  
(middle)**



Score: 0.60

**Floor 10  
(late)**



Score: 0.60

## Chapter 11

# Activation Patching

*Brain Surgery on the Robot*

### If I were 5...

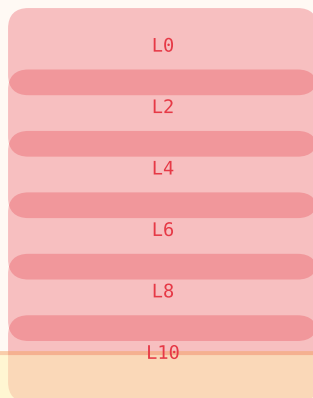
This is the coolest part! We do BRAIN SURGERY on the robot (don't worry, it doesn't hurt). We ask the robot two questions at the same time: a math question and a poetry question. Then we SWAP a piece of the math brain into the poetry brain at different floors and see what happens to the answer! If swapping at floor 11 makes the poem turn into math, we know floor 11 is where the "what to say" decision lives. It's like swapping ingredients in a recipe to find which one makes it spicy!

### The real deal:

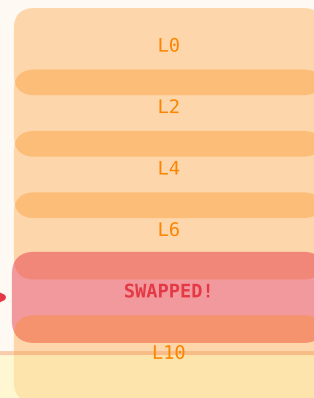
Activation patching: run source (math) and target (creative) prompts, cache all layer activations. Replace the target's residual stream at layer L with the source's. Measure KL divergence between patched output and clean target output. Lower KL = layer L carries more causal information about what the model will output. We sweep all 12 layers. Result: monotonically decreasing KL from layer 0 to 11. Layer 11 (KL~0) is most causally important.

### Brain Surgery: Swapping Pieces Between Runs

#### Math Brain



#### Creative Brain



SWAP!

#### What happens to the output?

If swapping at floor 11 changes the poem into math-like output --> that floor MATTERS.

# Surgical vs. Full Brain Swap

*The distributed nature of knowledge*

## If I were 5...

We tried two kinds of surgery: (1) swapping the ENTIRE floor (all 768 slots), and (2) swapping just the 20 most important drawers. Guess what? Swapping the whole floor worked great -- the poem started to look like math! But swapping just 20 drawers barely did anything. This means the "math knowledge" isn't hiding in just a few special drawers -- it's SPREAD OUT across hundreds of drawers, like butter on toast!

## The real deal:

Full residual patch at layer 11: KL = 0.75 (strong shift). Surgical patch of top-20 discriminative feature directions: KL = 3.38 (baseline = 3.54, so only a 0.16 reduction). The top-20 features account for ~6% of the total causal effect. Conclusion: math-relevant information is distributed across many features in superposition, not concentrated in a few 'math neurons'. This is consistent with the superposition hypothesis.

## Full Swap

(entire floor)

All 768 dimensions swapped  
KL = 0.75 (BIG change!)

## Surgical Swap

(top 20 features only)

Only 20 of 24,576 features  
KL = 3.38 (tiny change...)

## The Big Lesson:

Math knowledge isn't in 20 special drawers. It's SPREAD across hundreds.

*Like butter on toast -- you can't pick out just the butter.*

(This is what 'superposition' looks like in practice.)

# Feature Steering

*Pushing the Robot's Thoughts*

### **If I were 5...**

What if you could push the robot's thoughts in a direction? Like, the robot is thinking about poetry, but you sneak in and turn the "math" dial up to 5x! Will the robot start writing math instead of poems? We tried this! We amplified the 20 strongest math drawers by 5 times while the robot was thinking about a creative prompt. The output changed a bit -- but didn't become clearly math-like. The math thoughts are too spread out to push with just 20 drawers.

### **The real deal:**

Feature steering: during generation, we add 5x the top-20 math feature directions to the residual stream at the middle layer. This modifies the model's internal state to amplify math-related concepts. Result: output was altered but not clearly math-like, confirming the distributed nature finding from patching. More aggressive steering (100+ features, multi-layer, higher multiplier) would be needed for stronger effects.

### **Turning the Math Dial While Thinking About Poetry**

#### **Before Steering:**

Input: "Write a poem"  
**Output: beautiful poem**

Math dial  
5x!



#### **After Steering:**

Input: "Write a poem"  
**Output: slightly weird poem**

#### **Why didn't it become math?**

Because 20 features out of 24,576 is like turning up 20 pixels on a big screen -- the picture barely changes. You'd need hundreds of features to really shift the output.

# What We Discovered

*The Big Answers*

### YES, different questions light up different drawers!

Each prompt category has a distinct feature activation signature. The model genuinely processes math differently from poetry at the feature level.

### The robot figures out the question type EARLY!

Layer 2 (floor 2) has the best separability (Fisher = 1.33). The model categorizes inputs quickly, then blends for generation.

### Creative thinking is BROAD, math thinking is NARROW!

Creative L0 = 332 features vs. math L0 = 15. Creative writing activates 22x more features than math -- broader conceptual demands.

### Knowledge is SPREAD like butter, not in one jar!

Surgical patching of top-20 features captured only 6% of the effect. Information is distributed across hundreds of features in superposition.

### Facts have the strongest single feature!

Feature #6222 has selectivity score 10.25 for factual prompts -- the single most category-specific feature. Factual recall may be more localized than other capabilities.

### Later floors make the final decisions!

Layer 11 is most causally important (KL  $\rightarrow$  0 when patched). The model makes output decisions in the final layers.



## Quick Reference

# The Friendly Glossary

*Every big word, explained simply*

| Term                | Friendly Version                    | Expert Version                                                  |
|---------------------|-------------------------------------|-----------------------------------------------------------------|
| LLM                 | <i>A word-guessing robot</i>        | Large Language Model -- predicts next tokens autoregressively   |
| Layer               | <i>A floor in the building</i>      | One transformer block (attention + MLP + residual connection)   |
| Residual Stream     | <i>The hallway between floors</i>   | 768-dim vector carrying accumulated information                 |
| Superposition       | <i>Too many toys in one box</i>     | Encoding more features than dimensions via near-orthogonality   |
| SAE                 | <i>The magic sorting machine</i>    | Sparse Autoencoder -- decomposes 768 dims into 24,576 features  |
| Feature             | <i>One labeled drawer</i>           | A monosemantic direction in activation space                    |
| L0 (Sparsity)       | <i>How many lights are on</i>       | Count of nonzero SAE feature activations                        |
| Cosine Similarity   | <i>Pattern overlap</i>              | Dot product of normalized vectors; measures direction alignment |
| Fisher Ratio        | <i>How easy to tell apart</i>       | Between-class / within-class variance                           |
| Activation Patching | <i>Brain surgery</i>                | Swap activations between runs to measure causal importance      |
| KL Divergence       | <i>How different the answer got</i> | Information-theoretic measure of distribution difference        |
| Feature Steering    | <i>Pushing the dial</i>             | Amplifying feature directions to steer model output             |

*Now you know what's happening*  
**inside the AI's brain!**

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Whether you're five or fifty,  
the ideas are the same --

**just the words get bigger.**

Model: GPT2 | SAE: gpt2-small-res-jb | 24,576 features

CS 463 -- Machine Learning -- University of San Francisco -- Spring 2026